TRACE-RX Training Proposal: Critique

Review of the TikTok TechJam 2026 training proposal (Transformation-Reliability-Aware AI Image Detection). Prepared 30 August 2026.

Overall assessment

This is a strong, unusually literature-aware proposal. The core thesis — measure whether a forensic signal survived rather than letting every feature vote — is the correct response to the robustness findings in the detection literature, and the lineage-safe splits, low-FPR focus, abstention region and locked-test discipline are things most hackathon teams get wrong. The problems are mostly about **scope versus 24 hours** and about the two most novel components being **under-specified**. Issues below are listed roughly in order of importance.

Issue summary

#	Issue	Severity	Fix in one line
1	Scope exceeds a 24-hour, two-Spark budget	High	Define a minimum viable TRACE-RX due by hour 12; everything else is stretch
2	Availability and fusion formula under-specified	High	Supervise availability from known transform lineage; make logistic stacking the default
3	Selection metrics below the noise floor	High	Fewer decisions from dev data; loosen 0.5-point removal rule; add seeds for finalists
4	Data sourcing and format shortcut unstated	High	State master sources and canonicalisation; run metadata-only baseline before hour 5
5	Frozen backbones leave the biggest win unclaimed	Medium	Add one LoRA/adapter run on the chosen backbone before A1
6	From-scratch forensic tower likely undertrained	Medium	Make the pretrained fallback the primary; cut phase and neighbour modalities
7	Prior art (RIGID) not positioned against	Medium	Cite and state the delta explicitly
8	Assorted inconsistencies (objectives, abstention, provenance, licence)	Low	See section 8

1. It is a research programme, not a 24-hour plan

Count what has to land on two DGX Sparks: caching three ViT-L backbones over 152k records; training a 40–60M five-modality multi-scale transformer from scratch; a three-backbone trial; a four-rung component ladder (A0–A3); four objective variants (O0–O3); greedy intervention selection; a 16-condition transform sweep; a fractional-factorial interaction study; order-effect and saturation curves; and a fusion layer with learned availability, heteroscedastic variance and low-rank covariance.

A DGX Spark is roughly a mid-range consumer GPU with a large unified memory pool; it is not fast. Everything scientifically interesting (A3, the covariance fusion, CVaR/pAUC) is scheduled after hour 9 with zero slack, and the kill order only trims the periphery. If the forensic tower slips by two hours, the entire novelty of the proposal is at risk.

Recommendation. Define a minimum viable TRACE-RX — one backbone, one forensic branch, logistic stacking, one or two interventions — that must be finished and evaluated by hour 12. Treat the rest as stretch goals with explicit entry conditions.

2. The novel parts are the least specified

Availability a_k

This is the proposal's contribution and it has no training target. What supervises g_k? The answer is already in the data design: every record has a known transform lineage, so per-branch survival $\rho(k,t)$ can be computed on development and used (or thresholded) as the availability target. State this explicitly. Also flag the shortcut risk: a_k can trivially learn "this image has JPEG blocking" and become a codec detector rather than a signal-survival estimator, which reintroduces the format shortcut the proposal is trying to remove.

The fusion formula

$w = \Sigma^{-1}a / (a^T\Sigma^{-1}a + \varepsilon)$ is a GLS/BLUE-style combination. It assumes the experts are unbiased estimates of the same quantity on a common scale with error covariance Σ. Raw logits from different heads are not on a common scale; weights can go negative once off-diagonal terms enter; and Σ is being fitted across K experts plus intervention-response features on only 8,000 records. Make logistic stacking the default and the covariance layer the ablation, not the reverse.

Intervention responses at inference

Each intervention re-runs the relevant expert, so five operators across two experts is roughly a 6× latency multiplier. The utility function U(P) accounts for this, but the proposal should state the target latency budget up front so the greedy selector has a real constraint.

3. Statistical power does not support the selection machinery

- Development contains about 1,000 authentic masters. TPR@1%FPR is decided by 10 images, yet it carries 0.35 of the selection score and drives greedy intervention selection. The selector will fit noise.
- The removal rule "drop a branch with less than 0.5 point fused gain" is below single-seed variance, and screening is committed to one seed. The rule cannot distinguish signal from noise.
- Worst-group AUC over generator × source × journey groups on 4,000 dev records means some groups contain a few dozen images.
- The test-set caveat (one false positive = 0.2 pp FPR) is correctly stated for the locked set but the same arithmetic applies to development and is not acknowledged.

Recommendation. Reduce the number of decisions made from development data, loosen the thresholds to something above seed variance, and run at least three seeds on the finalist rather than one.

4. Data provenance is the biggest unstated risk

The proposal never says where 40,000 authentic and 40,000 AI masters come from, or how authentic subtype coverage (screenshots, scans, art, CGI) is obtained. If masters are pulled from GenImage-style sources, the JPEG-real / PNG-fake bias the proposal explicitly worries about is already present, and "one journey per master" does not fix the *clean* record.

- State the canonicalisation rule: decode everything and re-encode to a matched format, or train only on transformed views.
- Run the metadata-only baseline before hour 5 as a gate, not during error analysis.
- One fixed journey endpoint per master means each image is seen with a single augmentation for the whole run. On-the-fly augmentation is cheap and strictly better for the experts; reserve fixed records for evaluation.
- The journey sampling distribution is unspecified. It should mirror the official transform set and severities.

5. Frozen backbones probably leave the biggest win on the table

The literature the proposal cites (DRCT, FatFormer) shows that tuning the backbone beats a frozen linear head specifically on post-processing robustness. A one-hour LoRA or adapter run on the chosen backbone is feasible on a Spark and is likely to yield more than the entire covariance-fusion apparatus. Add it as an experiment, ideally before A1, and compare against the frozen probe at low FPR.

6. Forensic tower: start with the fallback

A 40–60M multi-modal transformer trained from scratch in roughly four hours on 144k images will be undertrained. The stated fallback (a pretrained no-early-downsampling RGB/residual network in the Gragnaniello style) is the more likely finisher. Make it the primary and the custom tower the stretch.

Of the five modalities, Fourier phase and neighbour relations have weak support: phase is destroyed by any shift or crop, and NPR-style features collapse under JPEG. They mostly exist to be "unavailable", which is conceptually consistent with the thesis but costs hours. RGB + residual + DCT is enough to test the idea.

7. Prior art not positioned against

The "response to an imperceptible perturbation" idea is essentially RIGID (He et al., 2024 — training-free detection via DINOv2 feature sensitivity to noise) and the broader test-time-consistency family. Cite it and state precisely what is different: learned availability from responses, multiple operators, and responses used as fusion features rather than a standalone score. Otherwise a reviewer will read it as a rediscovery.

8. Smaller issues

- **Objectives contradiction.** Section 10 says experts are trained once and objective sweeps run as fusion experiments, but CVaR and pAUC shape the expert representation. Clarify that O0–O3 apply to heads on cached features, and accept that this tests less than the text implies.
- **Abstention versus scoring.** If organiser evaluation is AUC/TPR over all images, "insufficient evidence" must still emit a probability. Treat abstention as a reporting layer, not an output class.
- **Provenance channel.** C2PA and watermark checks appear in the architecture but there is no implementation and competition data will not carry them. Mark as out of scope or remove from the diagram.
- **Screenshot and recapture** appear in the problem statement but not in the transform set. State that they are out of scope.
- **Selection score.** 0.10·(1–ECE) and 0.10·(1–worst-real FPR) are near-constant terms; the score is effectively 0.35 TPR@1% + 0.25 worst-group + 0.20 unseen. Say so.
- **Licensing.** DINOv3's gated licence may conflict with the competition's open-model rule; verify before caching features.
- **PE-SPC citation** (arXiv 2608.04935) could not be verified here. The proposal is right that 1.9B/6.7B results do not transfer to 300M; the mandatory ablation is the correct response.
- **Adversarial robustness** is never mentioned. It is reasonable to exclude, but say so in scope.

What to keep exactly as is

Lineage-safe partitions; the calibration and locked-test threshold protocol; the asymmetric treatment of missing evidence; the signal-survival $\rho(k,t)$ and interaction $I(a,b)$ definitions; and the error-analysis design. These are the strongest parts of the document and would read well even if half the architecture is cut.

Suggested minimum viable path

- **Hours 0–2:** manifests, canonicalisation, metadata-only baseline gate.
- **Hours 2–6:** cache features for all three backbones; train pretrained fallback forensic net with on-the-fly augmentation.
- **Hours 6–9:** backbone trial; one LoRA run on the winner; logistic stacking of global + forensic.
- **Hours 9–12:** two interventions (JPEG q95, blur 0.5) with lineage-supervised availability; calibrate; this is the deliverable.

- **Hours 12–20:** stretch in order — CVaR head, covariance layer, custom tower, additional interventions, interaction study.
- **Hours 20–24:** locked test once; report.